Prompt 1:

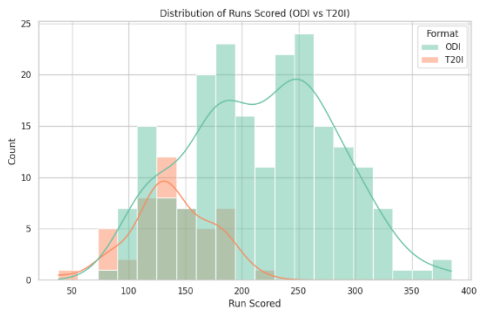
Act as a data analyst tasked with performing comprehensive exploratory data analysis (EDA) and visualization on a dataset. Begin by understanding the structure of the dataset: identify the number of rows and columns, data types, and key variables. Generate summary statistics (mean, median, standard deviation, min, max, quartiles) for numerical features and frequency distributions for categorical features. Highlight any inconsistencies such as missing values, duplicates, or outliers, and suggest appropriate handling strategies.

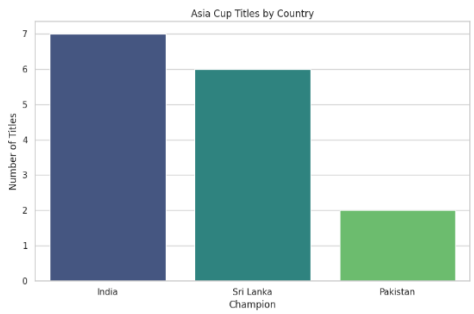
Create clear and insightful visualizations to uncover patterns and relationships. Use histograms and density plots to examine distributions of numerical variables, and boxplots to detect outliers. Generate bar charts or count plots for categorical variables to understand class balance. Use correlation matrices or heatmaps to identify relationships between numerical features, and pair plots for deeper multivariate analysis.

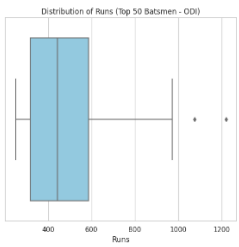
Where applicable, explore relationships between independent and target variables using scatter plots, grouped boxplots, or violin plots. Segment the data based on important categories to identify trends or anomalies across groups. If time-based data is present, perform time series analysis with line charts to identify seasonality or trends.

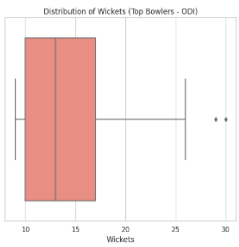
Interpret each visualization with clear, concise insights. Highlight key findings, unexpected patterns, or potential data quality issues. Provide recommendations for feature engineering, transformation (e.g., normalization, encoding), or further analysis steps. Ensure all visualizations are labeled properly with titles, axis labels, and legends, and use consistent color schemes for clarity.

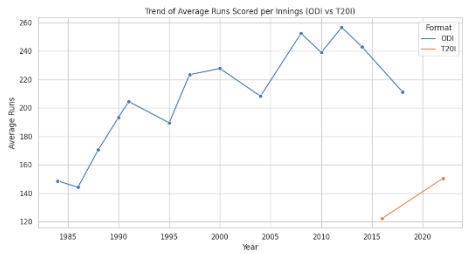
Conclude with a summary of the most important insights discovered during EDA and suggest next steps, such as modeling approaches or hypothesis testing. The goal is to turn raw data into actionable understanding through structured analysis and effective visualization.

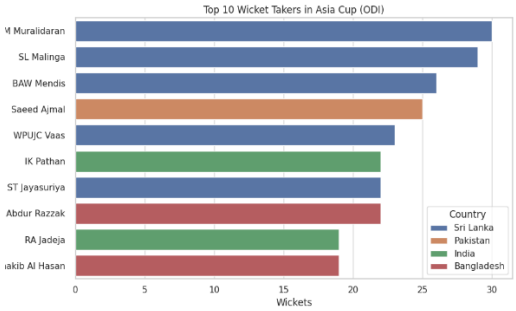












Prompt 2:

Act as a front-end developer and data visualization expert. Convert the results of my exploratory data analysis (EDA), including graphs, charts, and key insights, into a clean, responsive, and interactive webpage.

Use modern web technologies such as HTML, CSS, and JavaScript. For visualizations, use a suitable library like Chart.js, D3.js, or Plotly to render charts such as histograms, bar charts, scatter plots, boxplots, and heatmaps.

Structure the webpage into logical sections:

- A header with a title and brief dataset description
- A summary statistics section (tables or cards)
- A visualization dashboard displaying all graphs
- An insights section explaining key findings
- A conclusion or next steps section

Ensure the layout is visually appealing and user-friendly, using CSS (or frameworks like Bootstrap or Tailwind CSS). Make the page responsive so it works on both desktop and mobile devices.

Add interactivity where possible:

- Tooltips on charts
- Filters or dropdowns to explore subsets of data
- Hover effects and legends for clarity

Embed or load the dataset dynamically (e.g., from a CSV or JSON file), and ensure the charts update accordingly. Write clean, modular, and well-commented code.

Finally, ensure accessibility best practices (proper labels, contrast, semantic HTML) and optimize for performance.

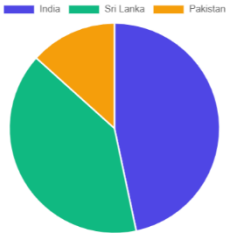
The goal is to transform static EDA outputs into an engaging, interactive web dashboard that communicates insights effectively.

Asia Cup Exploratory Data Analysis

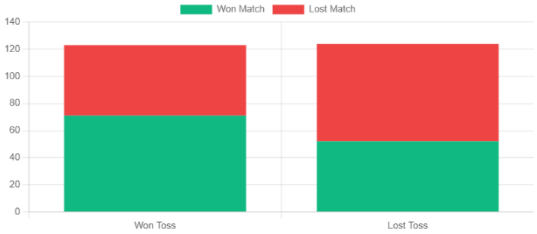
A comprehensive dashboard exploring four decades of cricket history, analyzing match trends, player performance, and tournament dominance.

TOTAL MATCHES	AVG RUNS / INNINGS	MOST SUCCESSFUL	MAX ODI SCORE
252	201.9	India	183

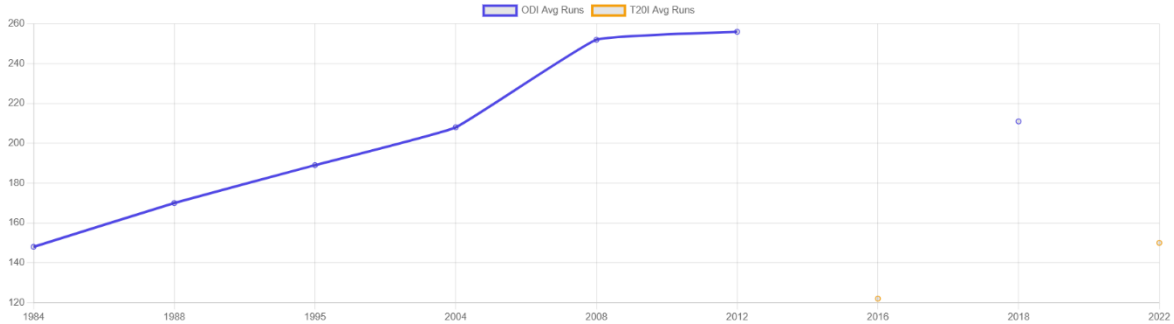
Asia Cup Titles by Country



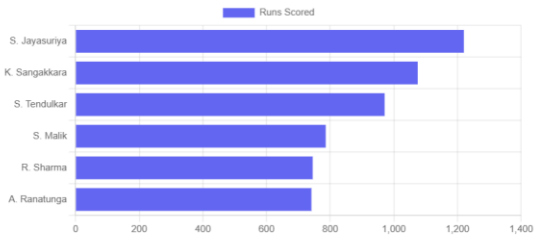
Toss vs Match Outcome



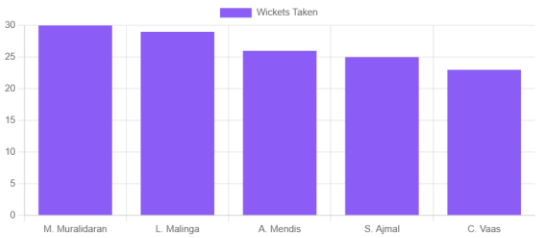
Average Runs Trend (ODI vs T20I)



All-Time Top 10 Run Scorers (ODI)



All-Time Top 10 Wicket Takers (ODI)



Key Insights

Dominant Forces

India and Sri Lanka have historically dominated the tournament, sharing 13 of the 15 titles. Their consistency across formats remains unmatched.

The Toss Factor

Winning the toss provides a psychological edge, but historical data shows a relatively balanced outcome, indicating high-quality competition regardless of batting order.

Rising Scoring Rates

The transition to T20I in 2016 and 2022 highlights a sharp increase in scoring intensity and run rates compared to traditional ODI formats in early years.